



School of Computer Science and Engineering

Deep Learning Project Review-2

Alzheimers Diseases Classification using cnn+adam

Guide

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Presented by

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ABSTRACT

This project focuses on the early detection of Alzheimer's disease using Convolutional Neural Networks (CNN) applied to MRI brain scan images. The model was trained on the Best Alzheimer MRI Dataset (99% Accuracy) from Kaggle, which contains four classes—Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Image preprocessing was performed to resize, normalize, and batch the data for efficient training. A deep CNN architecture was implemented with multiple convolutional, pooling, and dense layers to automatically extract spatial features from MRI images. The model was optimized using the Adam optimizer and trained for multiple epochs to achieve high accuracy. The dataset was divided into training and testing subsets to evaluate generalization performance. The trained model achieved strong classification accuracy, demonstrating its effectiveness in identifying various stages of Alzheimer's disease.

INTRODUCTION

- Focuses on early detection of Alzheimer's disease using MRI brain scans.
- Employs Convolutional Neural Networks (CNNs) for automatic image classification.
- Aims to improve diagnostic accuracy and reduce manual effort.
- Utilizes a Kaggle MRI dataset and the Adam optimization technique for model training.

PROBLEM STATEMENT

- Early detection of Alzheimer's disease is difficult due to subtle and overlapping brain changes.
- Manual MRI analysis by clinicians is labor-intensive and subject to human error.
- Lack of automated and accurate diagnostic systems delays timely treatment.
- A deep learning-based approach using CNNs can help classify Alzheimer's stages more reliably.

OBJECTIVES

- To preprocess and prepare MRI brain images for effective feature extraction and model training.
- To design and implement a Convolutional Neural Network (CNN) capable of classifying Alzheimer's disease into multiple stages — Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.
- To optimize the CNN model using the Adam optimizer for improved accuracy and faster convergence.
- To evaluate and visualize the model's performance through accuracy and loss plots, ensuring reliability for real-world medical applications.

LITERATURE REVIEW

Sl No.	Title, Year of Publication	Methodology	Dataset Details	Dis-Advantages	Results
1	A Classification based approach for early- stage Prediction and Multi-dimensional analysis of Alzheimer's Disease,2024	Inception v3 architecture	OASIS	Limited dataset, less generalization	developed a feasible approach, 96%
2	Alzheimer's Diseases Detection by using Convolution Neural Network, 2023	Convolution Neural Network	data set was created by several MRI scan images from various websites, including Kaggle and the National Institutes of Health (NIH)	Overfitting, small sample size	97%
3	Deep Learning Approaches for Early Detection of Alzheimer's Disease using MRI Neuroimaging, 2022	Deep Learning Models	ADNI	overfitting risk, high computation cost, limited interpretability.	81%
4	Deep Learning Based Hybrid Framework for Early-stage Prediction of Alzheimer Disease: A Predictive Analysis, 2025	Deep learning-based hybrid framework, ResNet50+ MobileNetV2 (ensemble)	OASIS	High computational cost	97% accuracy

LITERATURE REVIEW

SI No.	Title, Year of Publication	Methodology	Dataset Details	Dis-Advantages	Results
5	Early Detection of Alzheimer's Disease: An Extensive Review of Advancements in Machine Learning Mechanisms Using an Ensemble and Deep Learning Technique, 2023	Ensemble and Deep Learning Technique	ADNI	Review study only	KNN: 97%
6	Prediction and Classification of Alzheimer's Disease Using RESNET-152V2, 2024	ResNet-152v2 architecture	ADNI	Heavy model, needs GPU	92.5%
7	Revolutionizing Alzheimer's Disease Prediction Using EfficientNetB6, 2024	EfficientNetB6	ADNI	Complex architecture	97.78%
8	Revolutionizing Alzheimer's Disease Prediction using InceptionV3 in Deep Learning, 2023	InceptionV3	OASIS	Old model version	87.69%



LITERATURE REVIEW

SI No.	Title, Year of Publication	Methodology	Dataset Details	Dis-Advantages	Results
9	A deep learning framework for early diagnosis of Alzheimer's disease on MRI images, 2023	CNN, VGG16, ResNet50, DenseNet121	Kaggle Alzheimer's classification dataset	Limited generalization	99.98%, 97.44%
10	A Classification based approach for early stage Prediction and Multi-dimensional analysis of Alzheimer's Disease – 2024	Custom Inception CNN (modified InceptionV3 with ReLU, SoftMax)	OASIS MRI Dataset	Limited accuracy improvement, overfitting beyond 100 epochs	90.16% accuracy
11	A Survey on Machine Learning and Deep Learning Techniques for Alzheimer's Disease Prediction – 2023	CNN, SVM, ResNet, ANN (Survey)	ADNI and OASIS datasets	Review paper – no practical model tested	— (Survey results only)
12	Alzheimer's Disease Classification with a Hybrid CNN–SVM Approach on Enhanced MRI Data – 2024	Hybrid CNN + SVM (RBF kernel, Leaky ReLU)	MRI Dataset (6400 images, 4 classes: NoD, VMD, MiD, MoD)	Dataset dependency, possible overfitting	possible overfitting

LITERATURE REVIEW

SI No.	Title, Year of Publication	Methodology	Dataset Details	Dis-Advantages	Results
13	Alzheimer’s Disease Classification with Deep Learning Approaches – 2025	ConVNeXt + DINOv2 (CNN + Self-supervised model)	MRI + PET images (preprocessed, normalized, augmented)	High computational complexity, limited generalization	97.19% accuracy
14	Alzheimer’s Disease Detection Using Ensemble Techniques on MRI Images – 2025	Ensemble of EfficientNet-B0, Vision Transformer (ViT), and SqueezeNet	Kaggle MRI Dataset (4 stages of dementia)	High training cost, requires multi-GPU system	90.50% accuracy
15	Alzheimer’s Disease Detection Using MIC and MLP – 2023	MLP + Maximal Information Coefficient (MIC) + SMOTE	OASIS MRI + Neuropsychological test data	Imbalanced data, small sample size, moderate accuracy	88.4% accuracy

SOFTWARE AND HARDWARE REQUIREMENTS

SOFTWARE	HARDWARE
• Operating System: Windows 11 • Programming Language: Python 3.8+ • Libraries: TensorFlow, Keras, NumPy, Matplotlib, OpenCV • IDE: Google Colab	• Processor: Intel i5 • RAM: 16 GB recommended • Storage: Minimum 10 GB free space



PROPOSED SOLUTION

The proposed solution of this project is to build an efficient CNN-based image classification model that accurately identifies different brain image categories. The project focuses on preprocessing and standardizing image data to improve learning efficiency and applying deep learning techniques for better accuracy. The Adam optimizer is chosen for training because it combines the advantages of both AdaGrad and RMSProp, ensuring faster convergence and stable learning with adaptive learning rates. Ultimately, the project aims to support early detection and diagnosis through automated image analysis.

DATASET DETAILS

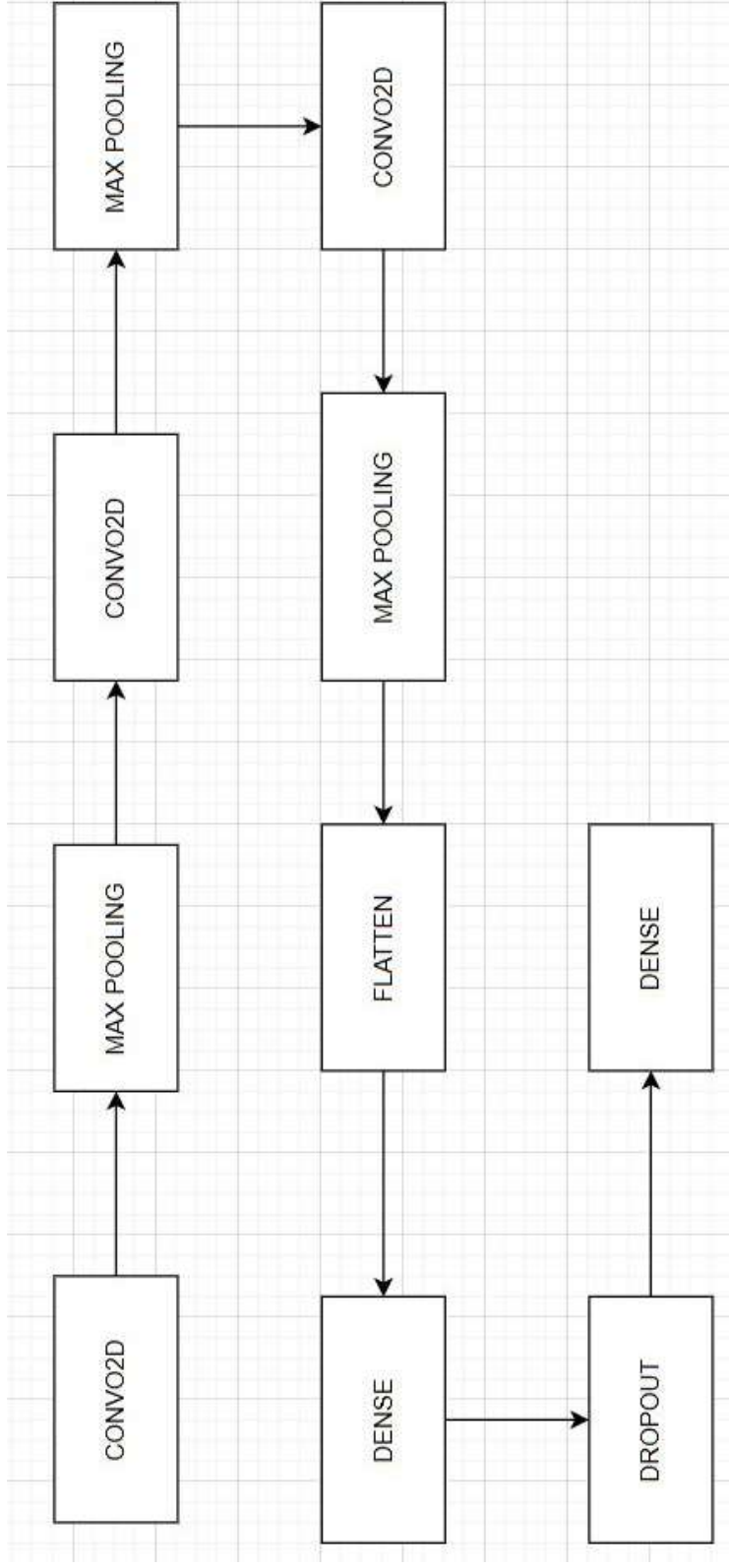
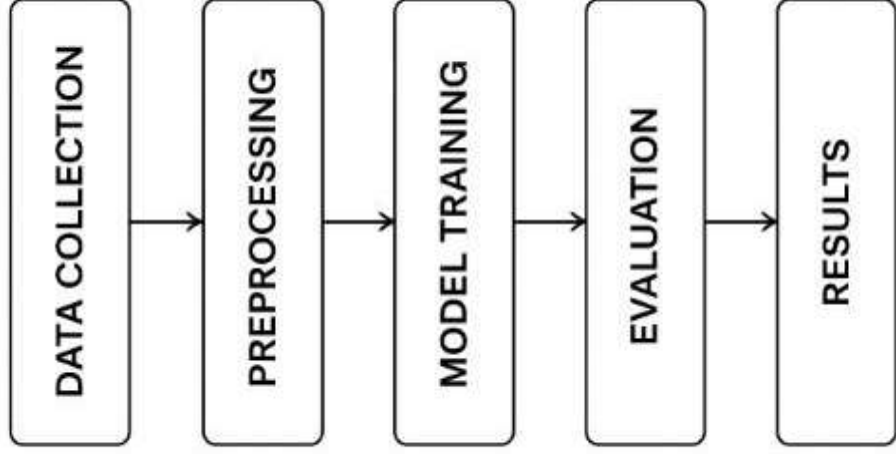
Best Alzheimer's MRI Dataset 99% Accuracy

SIZE: 75MB

SOURCE: KAGGLE

- Contains MRI brain scan images categorized into four distinct classes: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.
- Used in studies achieving **very high classification accuracy** (near or above 95% for multiclass use) when paired with CNN-based architectures.

METHODOLOGY



EXPLANATION

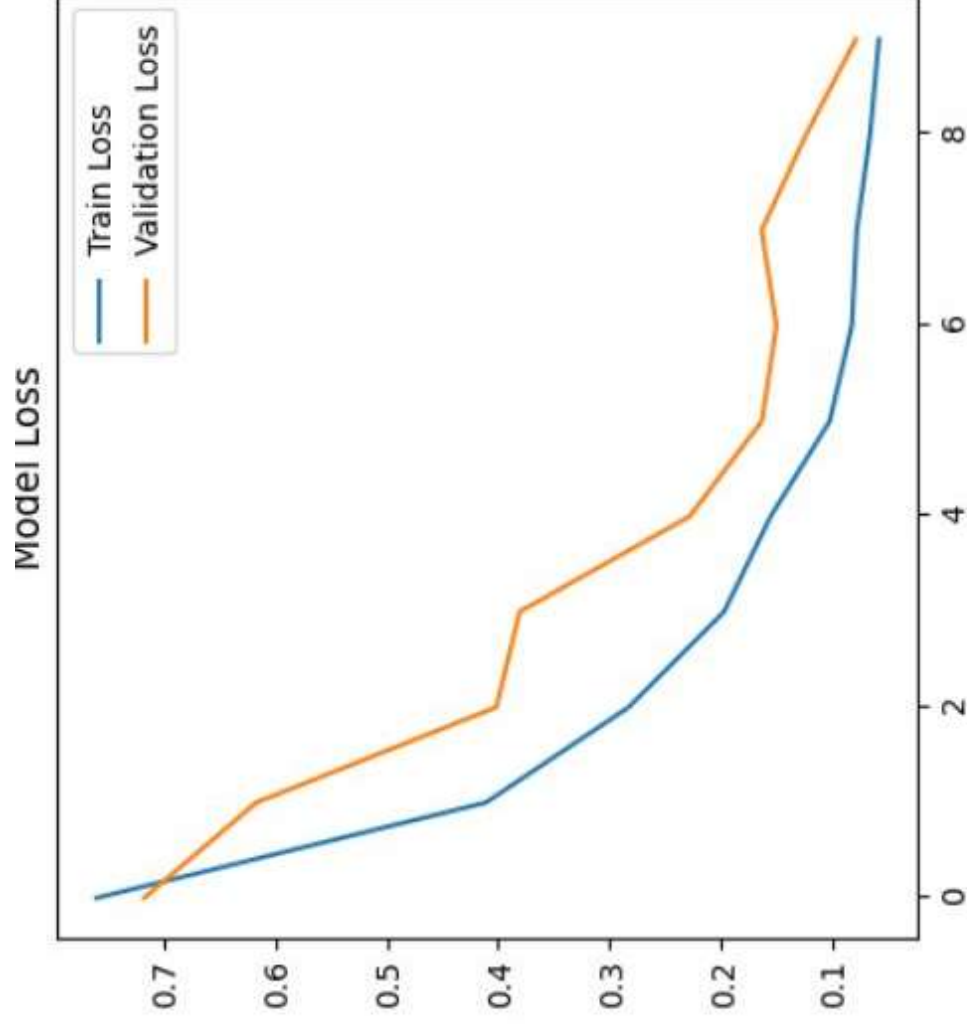
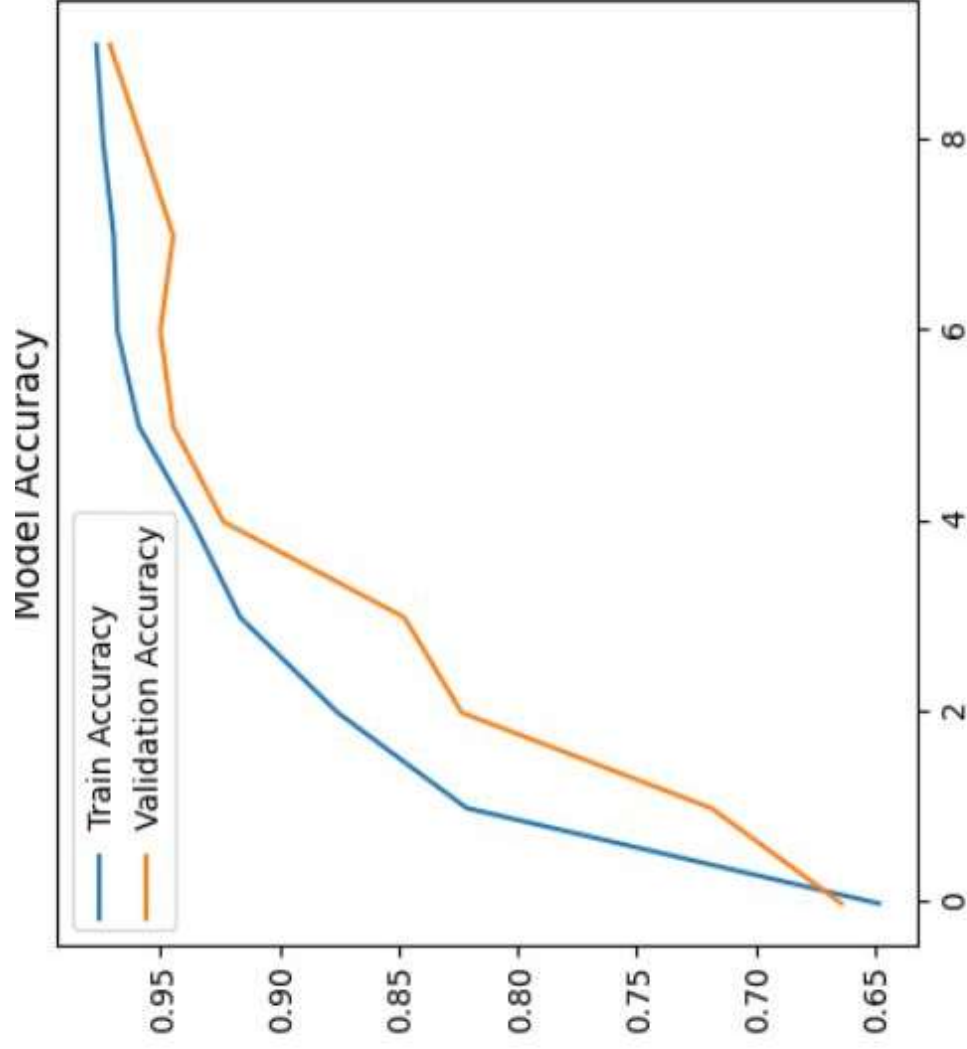
This project aims to develop an automated system for detecting Alzheimer's disease from MRI brain scans using Convolutional Neural Networks (CNNs). The model is trained on the Best Alzheimer's MRI Dataset (99% Accuracy), which includes four distinct categories — Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented — enabling accurate multiclass classification. Each MRI image undergoes preprocessing steps such as resizing, normalization, and data augmentation to enhance model performance and prevent overfitting.

The CNN model is designed to extract and learn essential spatial and structural patterns within brain tissues that indicate the progression of Alzheimer's disease. The Adam optimizer is employed for training due to its adaptive learning rate and efficient gradient handling, ensuring faster convergence and improved accuracy compared to traditional optimizers.

EXPLANATION

After training, the model's performance is evaluated through metrics like accuracy and loss curves to analyze its learning efficiency. The results demonstrate that deep learning techniques can significantly aid in medical image analysis by offering a **non-invasive, accurate, and efficient** method for early detection of Alzheimer's disease. This system has the potential to assist healthcare professionals in diagnosis and reduce human error in manual interpretation of MRI scans.

RESULTS



RESULTS

```
epochs = 10
history = model.fit(train_ds, validation_data=test_ds, epochs=epochs)
```

Epoch 1/10	320/320	300s	923ms/step	-	accuracy: 0.4982	-	loss: 1.0503	-	val_accuracy: 0.6638	-	val_loss: 0.7176
Epoch 2/10	320/320	298s	930ms/step	-	accuracy: 0.8094	-	loss: 0.4346	-	val_accuracy: 0.7185	-	val_loss: 0.6169
Epoch 3/10	320/320	296s	925ms/step	-	accuracy: 0.8690	-	loss: 0.2977	-	val_accuracy: 0.8233	-	val_loss: 0.4014
Epoch 4/10	320/320	294s	920ms/step	-	accuracy: 0.9100	-	loss: 0.2104	-	val_accuracy: 0.8475	-	val_loss: 0.3805
Epoch 5/10	320/320	294s	918ms/step	-	accuracy: 0.9310	-	loss: 0.1662	-	val_accuracy: 0.9234	-	val_loss: 0.2278
Epoch 6/10	320/320	288s	899ms/step	-	accuracy: 0.9618	-	loss: 0.1004	-	val_accuracy: 0.9445	-	val_loss: 0.1628
Epoch 7/10	320/320	290s	907ms/step	-	accuracy: 0.9695	-	loss: 0.0782	-	val_accuracy: 0.9500	-	val_loss: 0.1499
Epoch 8/10	320/320	325s	916ms/step	-	accuracy: 0.9698	-	loss: 0.0776	-	val_accuracy: 0.9445	-	val_loss: 0.1631
Epoch 9/10	320/320	292s	912ms/step	-	accuracy: 0.9748	-	loss: 0.0629	-	val_accuracy: 0.9578	-	val_loss: 0.1233
Epoch 10/10	320/320	291s	907ms/step	-	accuracy: 0.9779	-	loss: 0.0528	-	val_accuracy: 0.9711	-	val_loss: 0.0784

RESULTS

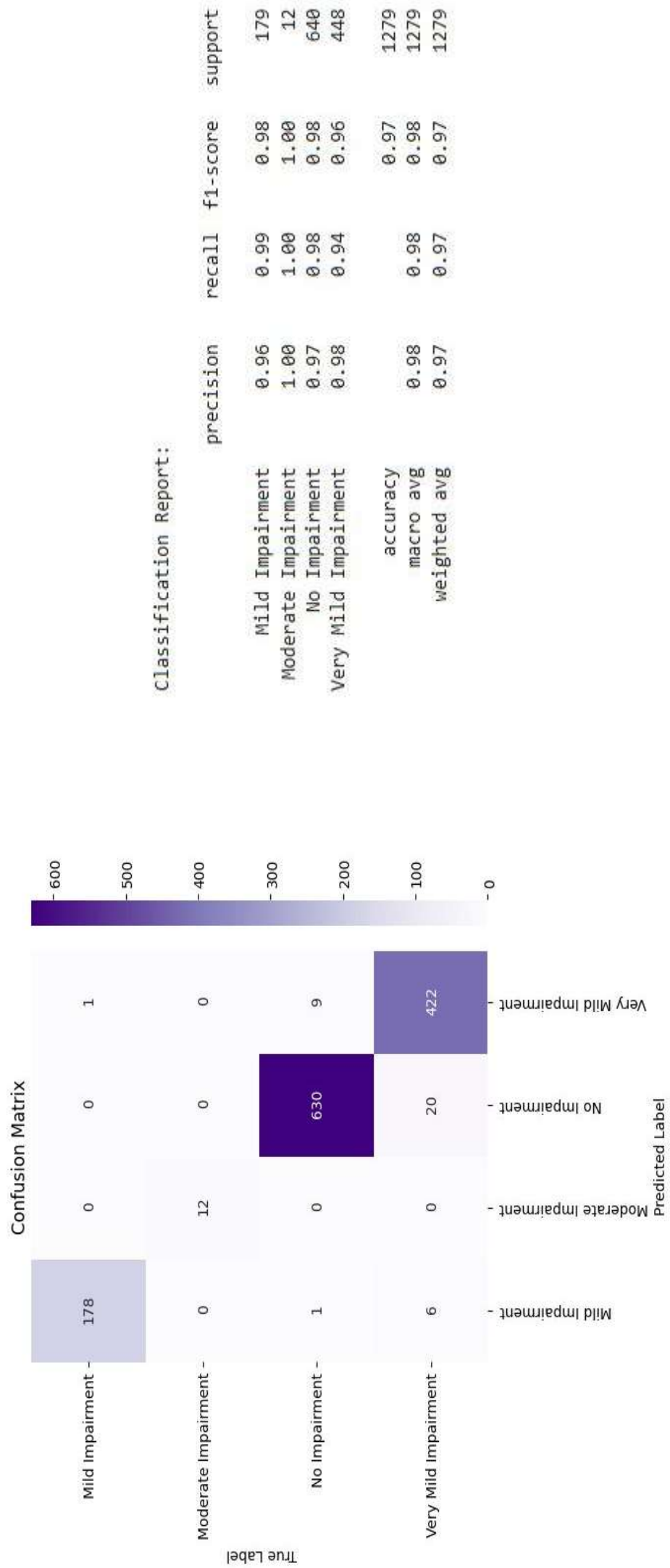
```
test_loss, test_acc = model.evaluate(test_ds)
print(f"\n📊 Test Accuracy: {test_acc*100:.2f}%")
```

```
40/40 ————— 10s 255ms/step - accuracy: 0.9854 - loss: 0.0526
```

📊 Test Accuracy: 97.11%



RESULTS



CONCLUSION

The proposed Alzheimer's disease detection system using Convolutional Neural Networks (CNNs) successfully classifies MRI brain images into different stages of dementia with 97% accuracy. By utilizing the Best Alzheimer's MRI Dataset (99% Accuracy) and the Adam optimizer, the model achieves efficient learning and reliable predictions. The results demonstrate that deep learning can effectively identify subtle structural changes in the brain, enabling early and accurate diagnosis of Alzheimer's disease. This approach reduces manual diagnostic errors and provides a faster, automated, and scalable solution for medical imaging analysis. Overall, the project highlights the potential of AI-powered healthcare systems in improving patient outcomes and supporting neurologists in clinical decision-making.

FUTURE ENHANCEMENTS

The trained CNN model can be deployed in a web or mobile-based healthcare application to enable real-time Alzheimer's screening. Users or clinicians can upload MRI images, and the system will instantly analyze them to predict the stage of Alzheimer's disease. This allows for quick, accessible, and early diagnosis, reducing the dependency on manual interpretation by specialists. Such integration can also assist doctors in monitoring patient progress and making timely treatment decisions.

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